

Bohdan Karavan

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EDUCATION

Brown University, 4.00/4.00 GPA

B.S. Computer Science, (full scholarship recipient)

Relevant Courses: Computer Graphics, Computer Networks, Computer Systems, Data Structures and Algorithms, Deep Learning, Data Science, Software Engineering, Functional Programming, Statistical Inference I, Multivariable Calculus, Linear Algebra

Providence, RI, USA

Aug 2022 - May 2026

WORK EXPERIENCE

Diia Ukraine, Ministry of Digital Transformation

Remote

Implementation Team Summer Intern

May 2024 – Aug 2024

- Redesigned the documentation for Diia, a government app with 21.5M+ users with 70+ official gov services available online
- Introduced an automated endpoint document decryption for API integration, improving efficiency of 15+ partner apps
- Researched and documented Diia's crucial services and architectures: Signature, Sharing, Validation, Authentication

Program Design with Data Structures and Algorithms

Providence, RI, USA

Teacher's Assistant for prof. Kathi Fisler, PhD

Jan 2024 – May 2024

- Owned the support cycle for an online education platform, claimed 200+ students' help tickets with a personalized response
- Led weekly office hours and hosted debugging sessions for the largest CS course at Brown, which has 450+ students enrolled
- Designed a systematic unit and integration test suite for "GraphQuest" assignment, uncovered a critical graph cycle bug

Integrated Introduction to Functional Programming

Providence, RI, USA

Teacher's Assistant for prof. John Hughes, PhD

Sep 2023 – Dec 2023

- Coached weekly debugging and conceptual office hours to help 120+ students engage with the topic of recursion
- Reviewed and planned weekly homework handouts, graded and gave personal feedback for 20+ projects
- Facilitated a weekly lab section with peer-to-peer code reviews, guided students through complex programming paradigms

CODING EXPERIENCE

Personal and School Projects

Sep 2022 – Present

- "Maps" project - created an interactable visualization of the US map in React based on historical redlining data from 1900s
- "Travel Buddy" - created an app to promote ride-sharing at Brown, with an ability to update ride preferences, search for destinations with a map from Google API, and store upcoming trips. Frontend written in TypeScript React, backend in Java
- "Movie Money Matters" - used Pandas, Random Forest Classifier and Hypothesis Testing to estimate movies' ratings based on its number of votes, duration, budget, and director. Based on datasets from IMDb and web-scraped data from Numbers
- Unified Artificial Language Transformers - implemented an LSTM model with Hyphenation, trained on Shakespearean English using TensorFlow, based on the Italian paper on Dante Alighieri. Currently working on expanding to other languages
- "Travel Planner" - utilizes Breadth-First-Search to find the shortest path between two cities and Dijkstra's Algorithm to find the most suitable path based on cost, time, or distance parameters given by user weights on any suitable graph
- Built a scalable multithreaded server that dynamically allocates threads based on resources in C that operates on a BST
- Created an AI player for a Connect4 game whose decision-making is based on depth 5 tree-like min-max algorithm

COMMUNITY SERVICE

Brown Entrepreneurship Program

Providence, RI, USA

Tech team lead

Jan 2024 – Present

- Led a team on designing and building the Brown Entrepreneurship Program website written with Next.js, React, and TailwindCSS. The website is now maintained, enabling a platform for the second largest club at Brown with 100+ members
- Conducted popular website research and delegated weekly team meetings. Led the discussion of low and high fidelity Figma designs, finalized the website template, and distributed the implementation tasks that led into production

Ukraine Global Scholars (UGS), 501(c)(3)

Ukraine, Remote

Participant & Mentor of a non-profit that prepares youth to get higher US education

Jun 2019 - Present

- Provided assistance to the nonprofit, which by now has helped 250+ Ukrainian students win \$62M+ in scholarships to top U.S. boarding schools and colleges. Participated as a mentor in the online pre-application bootcamp in 2020
- Led the application process of two students, helped lend two \$360k+ full scholarships at Yale and Stanford

SKILLS & INTERESTS

Technical Skills: Proficient in C, Python, Java, TypeScript, x86-64 asm, JavaScript, React, Next, ReasonML, HTML, CSS, SQL

Certificats: DeepLearning.AI TensorFlow Developer by DeepLearning.AI on Coursera. Certificate earned at September 2, 2023

Language: Ukrainian (native), Russian (fluent), English (fluent), German (intermediate), Japanese (beginner), Spanish (beginner)

Interests: Accomplished athlete in volleyball, tournaments won and participated in Ukraine, US; enthusiast in soccer, alpine skiing